GEOMETRIC REDUCEDNESS IN CHARACTERISTIC P VERSUS REDUCEDNESS OF THE FROBENIUS TWIST

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Allowed standard prerequisites (editorial).

- Geometric reducedness via the field of p-th roots. chapter lemma lemma-geometrically-reduced (not delivered as its own card). The criterion being applied.
- The Frobenius map on a field is an isomorphism onto its image. Algebra (field theory). Underlies the identification F_k = inclusion k -> k^{1/p}.

Source reference. Original lemma 035X: sources/stacks-chow/tags.txt, source lines 4144–4144. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field of characteristic $p > 0$. Let X be a scheme over k. Then X is geometrically reduced if and only if $X^{(p)}$ is reduced.*

Proof. Consider the absolute frobenius $F_k : k \to k$. Then $F_k(k) = k^p$ in other words, $F_k : k \to k$ is isomorphic to the embedding of k into $k^{1/p}$. Thus the lemma follows from Lemma 035X. □

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